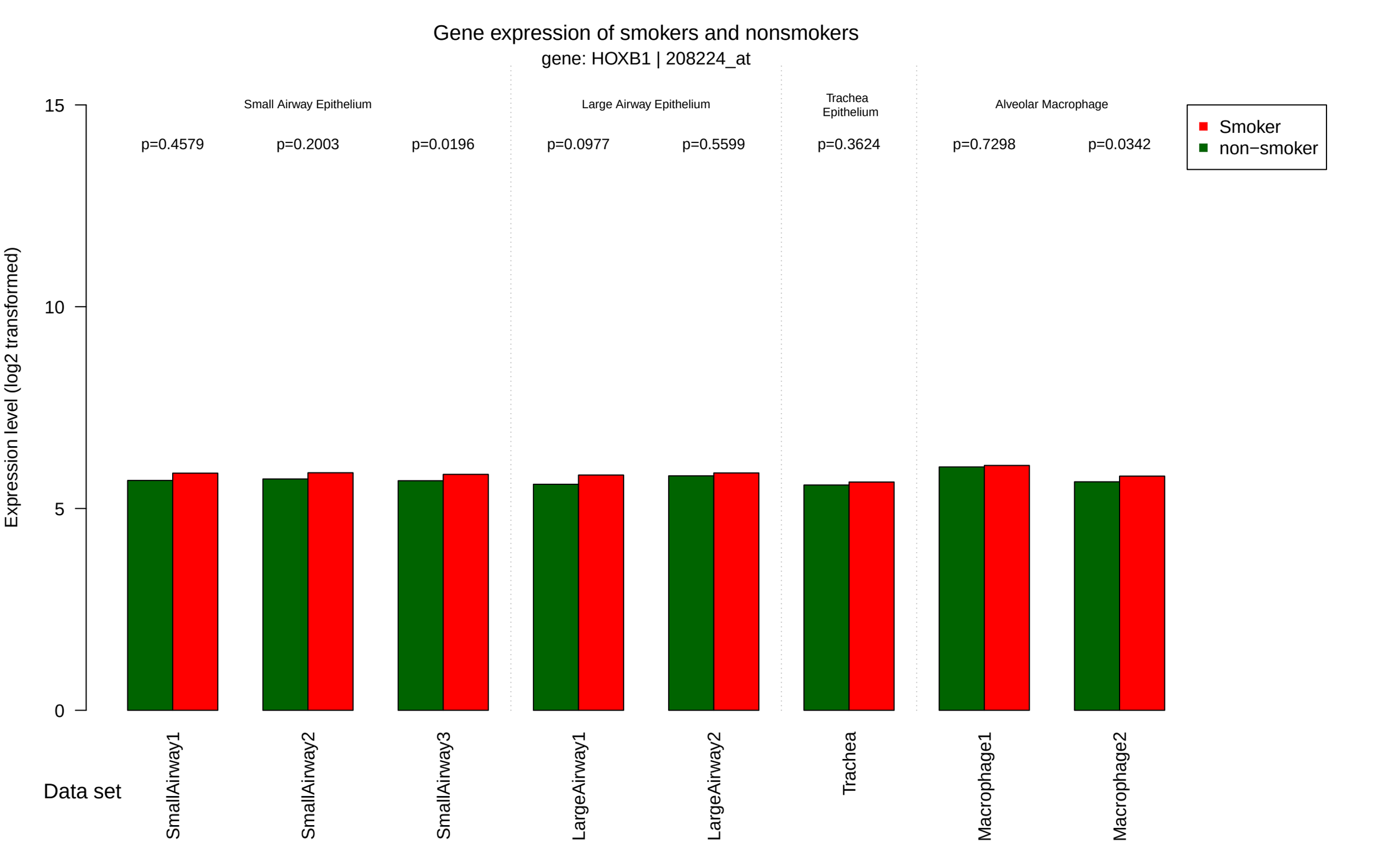
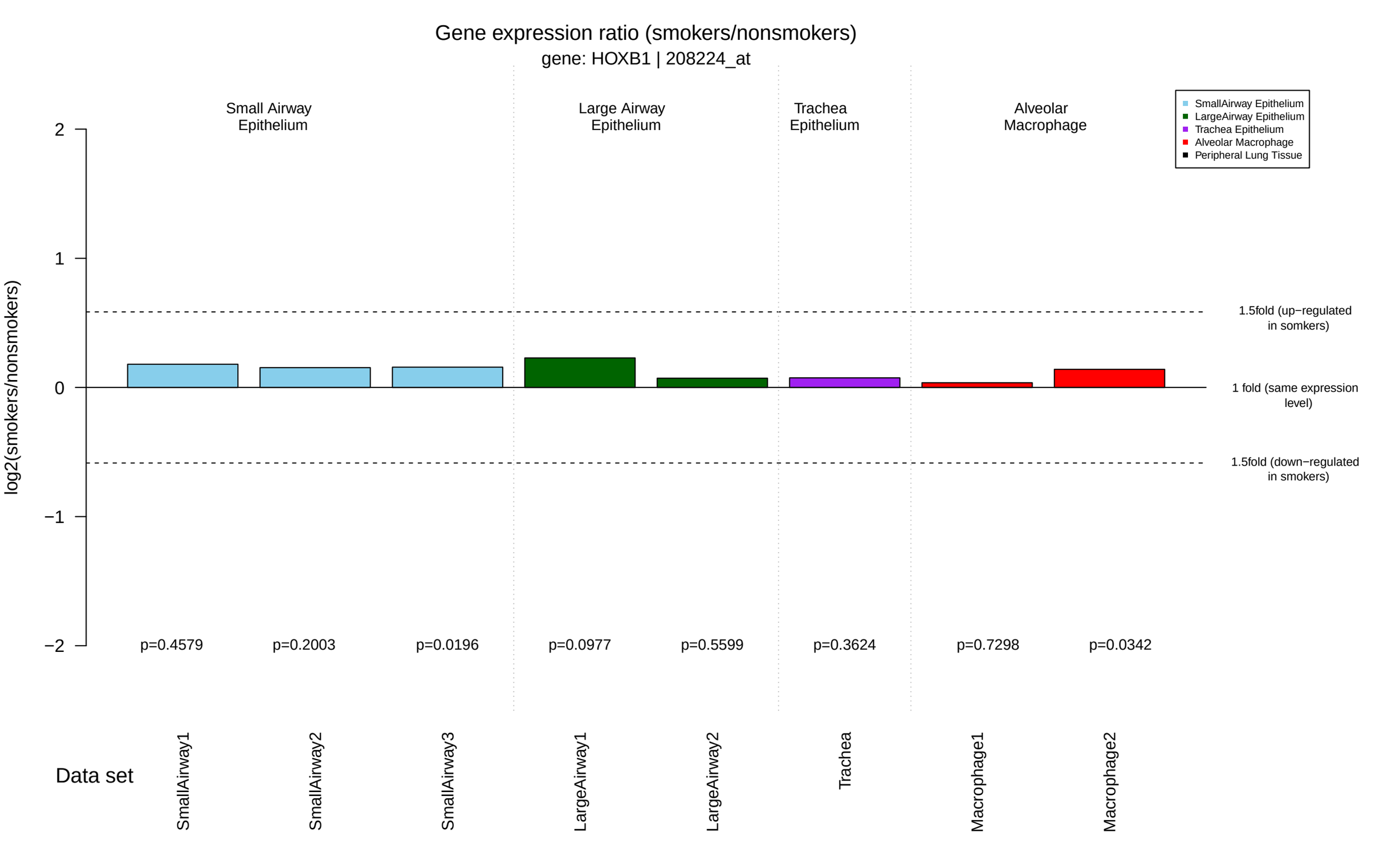
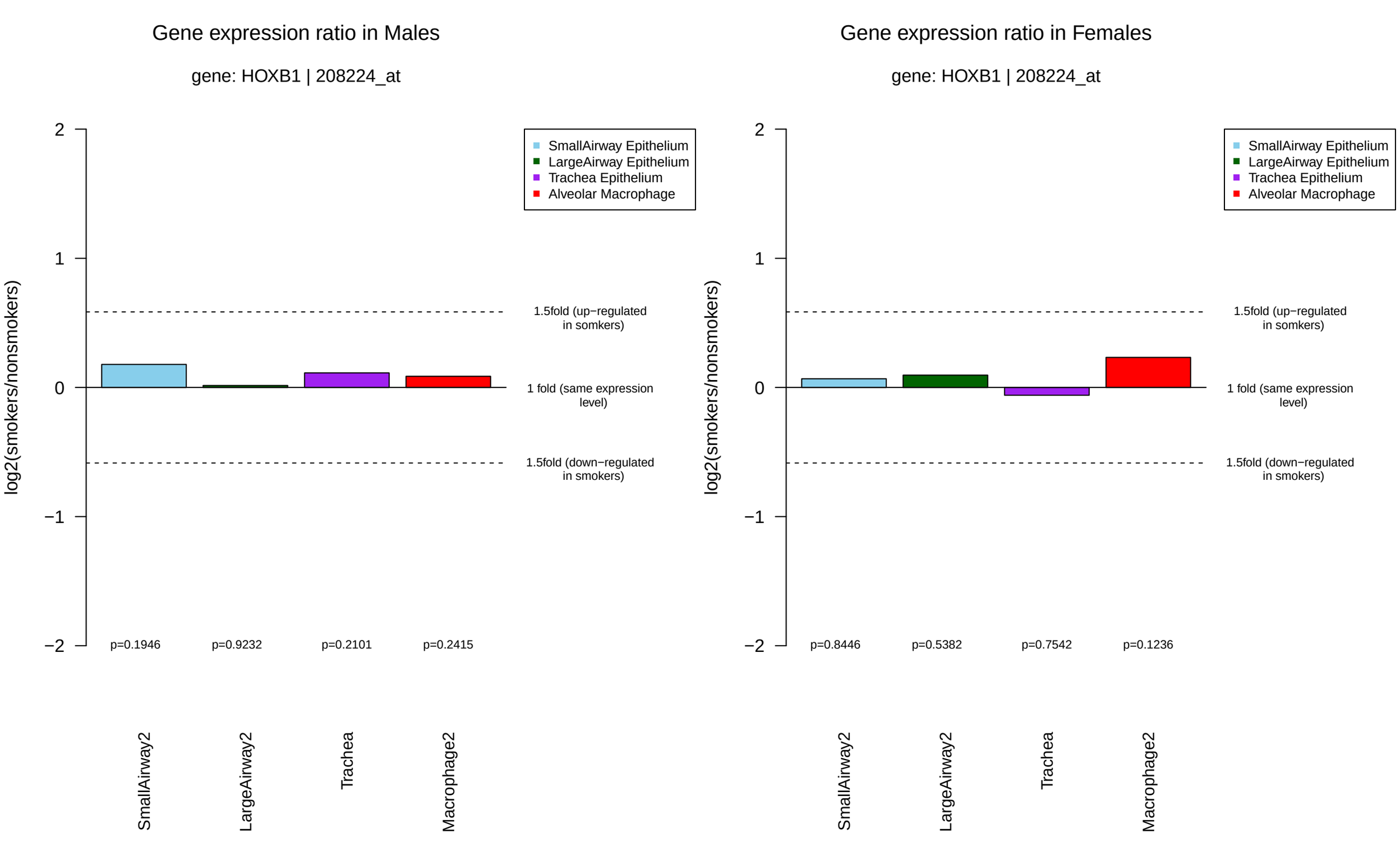
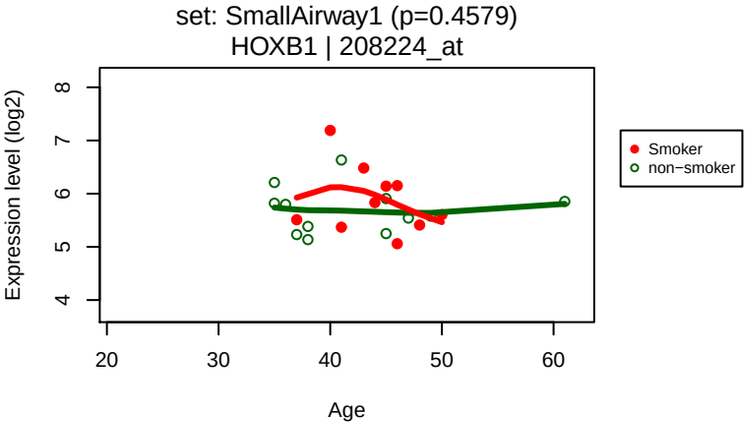
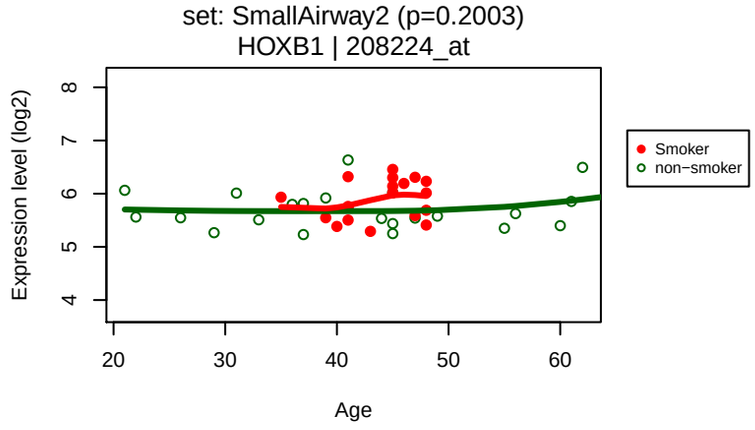
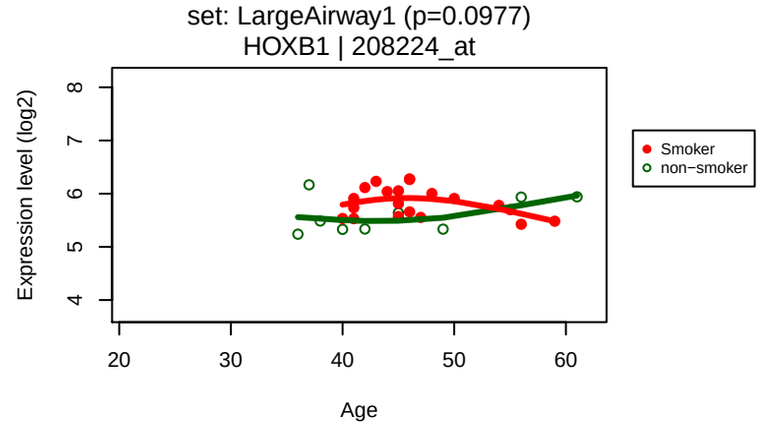

Gene expression of smokers and nonsmokers
gene: HOXB1 | 208224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4579
p=0.2003
p=0.0196
p=0.0977
p=0.5599
p=0.3624
p=0.7298
p=0.0342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

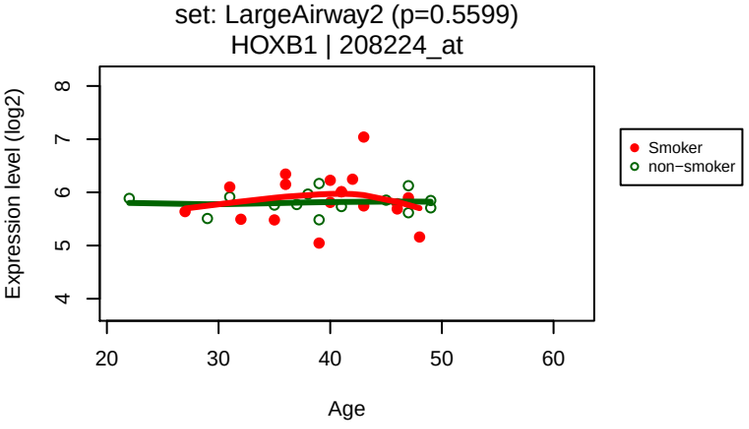

Gene expression ratio (smokers/nonsmokers)
gene: HOXB1 | 208224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4579
p=0.2003
p=0.0196
p=0.0977
p=0.5599
p=0.3624
p=0.7298
p=0.0342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

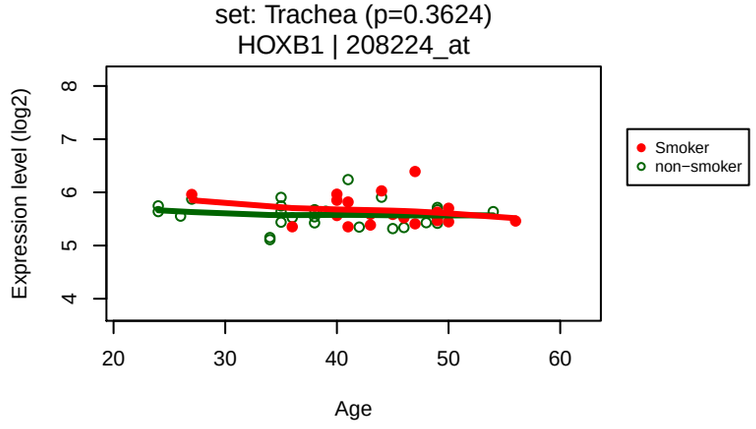

Gene expression ratio in Males
gene: HOXB1 | 208224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1946
p=0.9232
p=0.2101
p=0.2415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB1 | 208224_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8446
p=0.5382
p=0.7542
p=0.1236
SmallAirway2
LargeAirway2
Trachea
Macrophage2

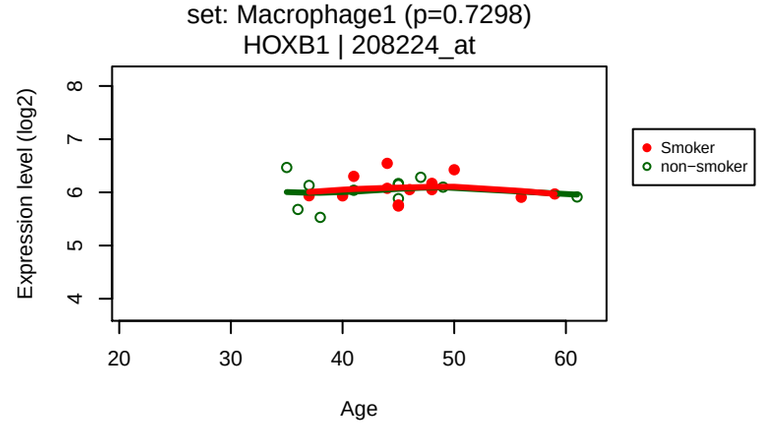

set: SmallAirway1 (p=0.4579)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker

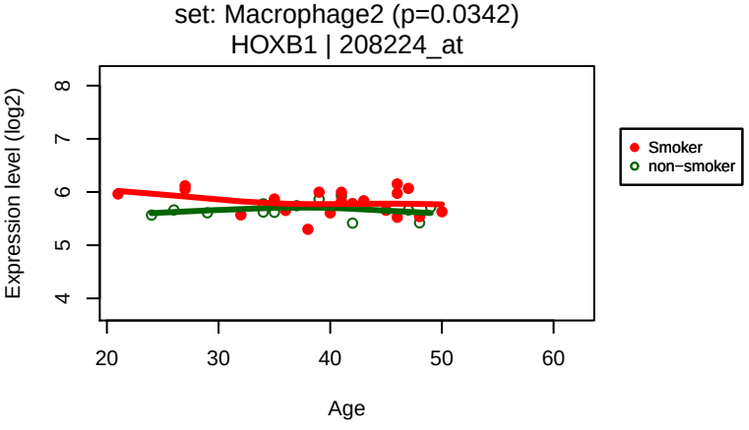

set: SmallAirway2 (p=0.2003)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker

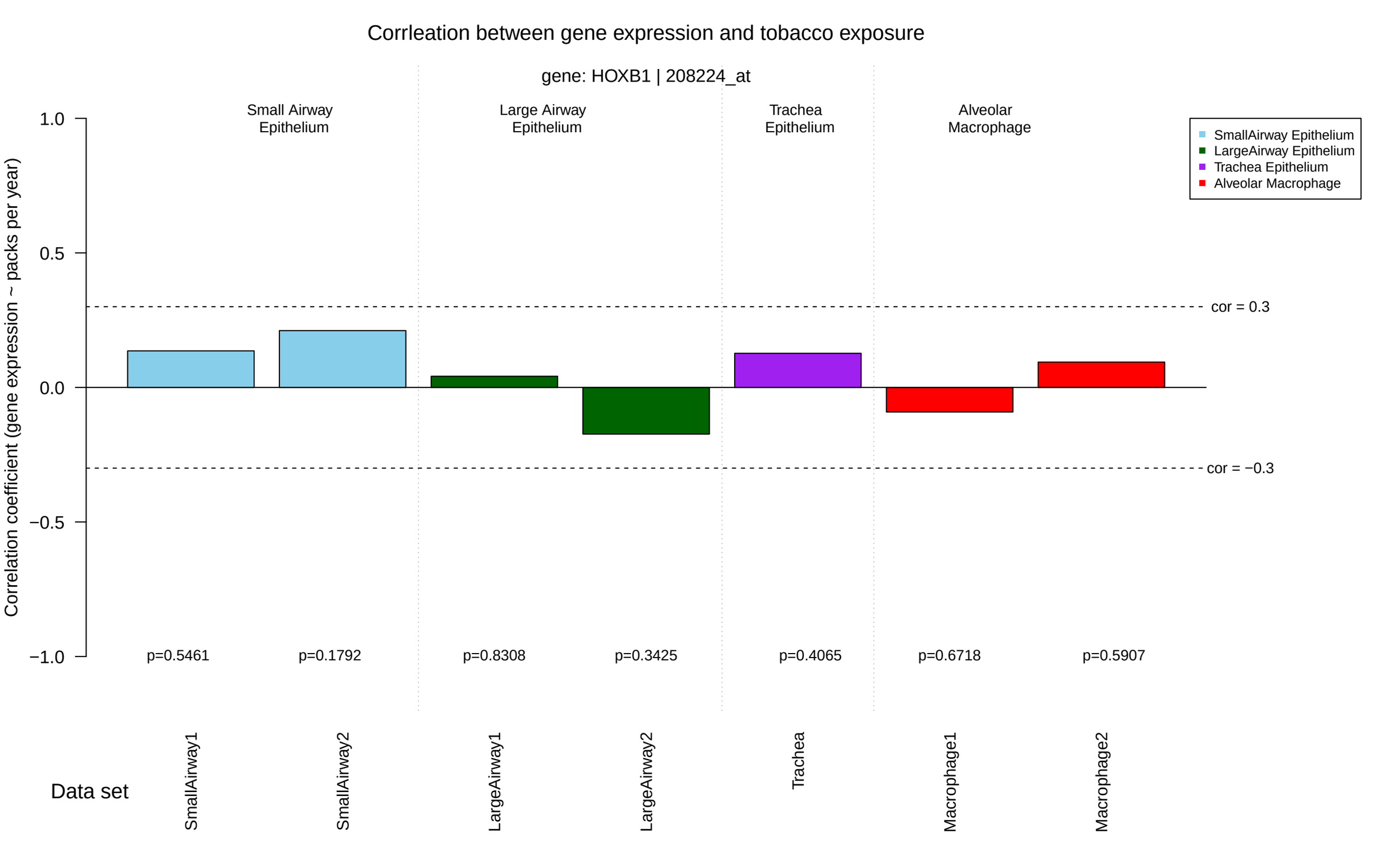

set: LargeAirway1 (p=0.0977)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5599)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3624)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7298)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0342)
HOXB1 | 208224_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB1 | 208224_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5461
p=0.1792
p=0.8308
p=0.3425
p=0.4065
p=0.6718
p=0.5907
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2